monotropism-scoring-levels-inertia

Scoring Monotropism: Inertia, "Friction," and Cognitive Energy — What the Evidence Actually Supports

A cited research brief — fact-checking the "scoring monotropism into levels" framing.

> ***Informational, not medical advice.*** *This brief evaluates how monotropism can and cannot be measured. The Monotropism Questionnaire (MQ) is a self-report research instrument,* ***not a diagnostic or clinical tool****, and it has no validated severity bands.*

Executive summary

- **Your instinct is right on the core idea and wrong on the packaging.** The notion that monotropism can be *measured* along a spectrum, and that the real quantity of interest is the **effort ("friction") of switching attentional states**, is well supported — but the specific four-level taxonomy ("Low / Moderate / High / Extreme-Absolute") is a popular framing, **not a validated scientific classification**.
- **Monotropism is correctly a *dimensional trait*, not a syndrome.** The only validated measure — the **Monotropism Questionnaire (MQ)**, Garau et al. 2023 — returns a continuous score from 1 to 5. It has **no published cutoff, no percentile bands, and no severity levels** [2]. Binning it into named levels is always an arbitrary editorial choice.
- **The "friction" of switching already has a name: *autistic inertia*.** This is a real, increasingly researched construct — difficulty *starting, stopping, and switching* tasks that is "not within conscious control." It

was first articulated in autistic community narratives and is now in peer-reviewed work [3][4][5]. Your "switching friction" is essentially this.

- **"Attentional inertia" conflates two distinct things.** *Autistic inertia* (task-level: hard to start/stop/switch) and *sticky attention / impaired disengagement* (stimulus-level: hard to pull attention off something, e.g. Landry & Bryson 2004; Sacrey 2015 meta-analysis) are related but measured differently [6][7]. Lumping them under one term loses precision.
- **"Cognitive energy economics" is the single strongest framing** — and it already exists academically as Goldknopf's **"atypical resource allocation"** model (2013), which treats autism as a difference in *how much and where* attentional resources are spent [10]. This is the neuroaffirming, non-deficit version of what you are describing.
- **The burnout/shutdown/meltdown link is real but heuristic.** Autistic burnout (exhaustion, loss of skills, reduced tolerance) is an established construct, and the *narrative* link to forced switching + masking is compelling and community-validated — but the specific pathway "monotropism → tunnel rupture → burnout" is largely theoretical and first-person, not yet experimentally established [8][9].

---

1. Can monotropism be "scored"? Yes — as a continuum, not as levels

Monotropism is an attention-based account of autism: a monotropic mind funnels processing resources into a small number of intensely-aroused "attention tunnels" rather than spreading them across many channels [1]. Because it is described as a *difference in resource allocation*, it is naturally a **spectrum**, not a category — which is exactly why your "levels" intuition feels right.

The validated instrument is the **Monotropism Questionnaire (MQ)** — 47 items, scored on a 5-point Likert scale, developed in 2023 by Garau and colleagues *in collaboration with an autistic-led community group* [2]. Two things are decisive for your question:

- **It returns a single continuous mean (1–5), plus eight subscale means.** In the validation study (1,110 participants), autistic participants averaged **4.15** and non-autistic participants **3.19**, with

ADHD independently raising scores [2]. The two distributions overlap heavily — many non-autistic people out-score some autistic people, and vice versa.

- **There is no cutoff, no diagnostic threshold, no severity band.** A scan of the validation paper confirms the word "cutoff" appears only as a *statistical* citation (Hu & Bentler's fit-index criteria) and "sensitivity" only as a *sensitivity analysis* of how autism was defined [2]. The MQ's own authors do not propose levels.

So what does a score "mean"? Only its relative position on the dimension. The community-built online app that inspired this question helpfully translates a score into a *percentile* ("more monotropic than about X% of autistic / allistic people"), computed from the validation sample — and that is a reasonable, honest way to locate a score. But it is **a descriptive percentile, not a clinical band**: "the 97th percentile of monotropic tendency" is meaningful; "Level 3 / Extreme monotropism" is a label the evidence does not confer.

> ***Bottom line:*** *score it, plot it, percentile it — but don't level it. The continuum is real; the named categories are not.*

2. Your "friction of switching" already exists: it's called *autistic inertia*

This is the most important correction-by-name in the brief. The phenomenon you describe — the cognitive cost of moving *into* and especially *out of* an attention tunnel — is an established construct called **autistic inertia**: a difficulty with **starting, stopping, and switching tasks that is not within conscious control** [3].

Three things make this directly relevant:

- **It originated in the autistic community and has entered the peer-reviewed literature.** The Association for Psychological Science describes autistic inertia as "a concept first articulated in autistic community narratives," now backed by qualitative and theoretical work [4]. Karen Buckle's first-hand-accounts study found participants repeatedly described being unable to start, stop, or change activities at will [3].

- **The literature literally uses your "friction" language.** The APS piece calls task initiation a "high-friction launch," frames switching as carrying "switching costs," and notes that once engaged, "switching focus requires more cognitive effort, sometimes leading to exhaustion" [4]. Your instinct to quantify *effort/friction* is the same move researchers are making.
- **It is now being formalised.** Recent peer-reviewed work treats the "stuck state" / limited response initiation as a multi-level phenomenon (muscular-behavioural "micro," relational and environmental "meso/macro") and co-creates support frameworks with autistic people [5]; others model it through predictive-processing / active-inference accounts (referenced in [4]).

In short: if you want to measure "switching friction," you are measuring autistic inertia — and that construct already has qualitative evidence, lived-experience validity, and a growing quantitative literature to build on.

3. "Attentional inertia" actually bundles two different mechanisms

Precision matters here, because the two mechanisms call for different language and (eventually) different measures.

3.1 *Autistic inertia* — task-level, about agency and momentum. Difficulty *initiating*, *terminating*, and *switching* activities, experienced as disproportionate effort and a loss of voluntary control (the "object in motion stays in motion / at rest stays at rest" feeling) [3][4]. This is the closest fit to your "pull themselves out of the tunnel" idea.

3.2 *Sticky attention / impaired disengagement* — stimulus-level, about perception. Difficulty *pulling attention away* from a stimulus once locked on. Landry & Bryson (2004) showed young autistic children were markedly slower to disengage from a central stimulus when a peripheral one appeared — "sticky attention" [6]. A 2015 meta-analysis confirmed slowed attention disengagement across studies [7], and the same sluggish disengagement appears in infants later diagnosed autistic [6]. This is measured with experimental tasks (gap/overlap paradigms), not questionnaires.

Why the distinction matters: sticky attention is a *perceptual* capture effect (you literally don't notice the interruption); autistic inertia is a *control/momentum* effect (you notice but can't shift). Your four-level descriptions actually slide between the two — e.g. "background stimuli are genuinely not processed by the brain" is sticky attention [6], whereas "forcible transitions feel like pulling a train off its tracks" is inertia [3]. A careful model should keep them labelled separately, because they may have different causes and different supports.

4. The four proposed "levels" — what's supported, what's invented

Rather than endorse or discard the taxonomy outright, here is each proposed band tested against the evidence.

Proposed level	Core claim	Evidence verdict
Low (polytropic-dominant)	Many "tabs" open; near-zero switching cost	**Supported as one end of the continuum.** The MQ's lower scorers and the non-autistic group mean (3.19) describe exactly this [2]. "Polytropic" is the theory's own term for it [1].
Moderate	Can focus deeply but retains baseline awareness; mild switching cost	**Supported as the middle of the dimension.** Flow states are reported across the population; the MQ scores between the group means capture this band [2][11]. But the *name* is arbitrary.
High	Preferred single intense tunnel; interruptions genuinely missed; high switching cost	**Mechanism supported, category not.** The components are real — missed interruptions = sticky attention [6][7]; high switching cost = autistic inertia [3][4]; strong autism/ADHD association = MQ findings [2]. But there is no threshold separating "High" from "Moderate."
Extreme / Absolute	Total immersion; disruption causes "neurological shock," meltdowns, shutdowns	**Overstated.** Deep immersion ("flow") and distress at disruption are real [11], but **"absolute" implies a qualitative jump the data don't show**, and **"neurological shock" is not a recognised term**. Meltdowns/shutdowns are involuntary overwhelm responses, not a "shock" specific to a monotropism band.

Three problems with publishing the taxonomy as-is:

1. **It implies categorical jumps where the data show a smooth continuum.** The MQ distributions of autistic and non-autistic people overlap substantially [2]; there is no natural "knee" that would justify four discrete levels.
2. **It smuggles in severity/pathology.** "Extreme/absolute … meltdowns, burnout" frames the high end as inherently harmful — the opposite of the neuroaffirming framing the theory intends [1]. Higher monotropism is *not* the same as worse outcomes; outcomes depend on *environmental fit*, not level.
3. **The bands have no reliability data.** A useful level system needs to show that people land in the same band on re-test and that bands predict something. No such study exists for monotropism.

What to do instead. Express the same idea as a *position on the dimension* plus a *profile*: "high monotropic tendency (≈top X% of the autistic sample), with pronounced autistic inertia on task-switching and frequent disruption-driven meltdowns in unsupportive environments." That is accurate, individualised, and non-pathologising.

5. Disruption → meltdown, shutdown, burnout: what the evidence supports

This part of your framing is the most consequential and needs the most care.

Meltdowns and shutdowns are real and well-described as involuntary responses to overwhelming stress or sensory/cognitive load (an externalising "meltdown" vs an internalising "shutdown" withdrawal) — recognised in clinical guidance and community consensus. They are triggered by *overwhelm*, of which tunnel-disruption is one common route [4][8].

Autistic burnout is now a defined construct. Two foundational studies — Raymaker et al. (2020) and Higgins et al. (2021, a Grounded-Delphi study with autistic experts by experience) — converge on a definition: **chronic exhaustion, loss of skills, and reduced tolerance to stimulus**, conceptualised as resulting from chronic life stress and "a mismatch of expectations and abilities" without adequate support and recovery [8][9].

The monotropism → burnout link is heuristically powerful but not yet experimentally established. The narrative is compelling and community-validated: every forced transition is a tunnel rupture; constant masking and switching in a polytropic world burns resources; without recovery/flow time, the system exhausts [8][9][11]. But the *specific* contribution of monotropic tunnel-disruption to burnout — versus masking, trauma, sensory load, and chronic demand — has not been isolated experimentally. State it as a *strong, testable hypothesis*, not an established fact.

> *The honest, evidence-based claim is: "Autistic burnout is driven by chronic stress, masking, and a mismatch between demands and capacity; the monotropism framework predicts that constant forced switching and tunnel-disruption are a major, under-recognised contributor — a prediction that fits lived experience and awaits direct testing."*

6. "Cognitive energy economics" — your strongest framing

Your closing move — shifting from "deficit" to "how much fuel a brain burns switching gears" — is exactly right, and it already has an academic home.

Goldknopf's **"atypical resource allocation"** model (2013) operationalises monotropism as a difference in *resource economics*: attentional resources in autism are **narrowed** (concentrated into fewer/smaller channels, sometimes more intensely) and/or **reduced in overall capacity** [10]. On this view, the "cost" of a transition is literally the energetic expense of breaking one allocation pattern and building another. That is "cognitive energy economics" stated precisely — and it is the framing most consistent with a non-pathologising account, because it frames the problem as *spending* (a difference in budgeting) rather than *brokenness*.

This is the framing to keep. Three advantages:

- It is **neutral about value**: intense focus is an asset when the environment supports it (flow, expertise) and a cost when the environment demands constant switching [10][11].

- It **predicts intervention**: reduce unnecessary transitions, protect flow/recovery time, match demands to capacity, and the "fuel bill" drops — which is precisely what burnout-prevention guidance recommends [8][9].
- It **unifies** the phenomena in your four levels without needing the levels: missed stimuli, switching cost, overwhelm, and burnout all become consequences of *how resources are allocated and depleted*, on a continuum.

7. Practical takeaways — what to claim, and what to avoid

Claim confidently:

- Monotropism is a measurable, **dimensional** trait (MQ, 1–5, with eight subscales); report the score and optional percentile, not a "level" [2].
- The **"switching friction"** you describe is **autistic inertia** — a real, community-originated, now-peer-reviewed construct [3][4][5].
- **Sticky attention / impaired disengagement** is a distinct, experimentally-measured mechanism [6][7].
- **Autistic burnout** is a defined construct driven by chronic stress, masking, and demand–capacity mismatch; monotropic tunnel-disruption is a *plausible, lived-experience-validated* contributor [8][9].
- "Cognitive energy economics" ≈ Goldknopf's **resource allocation** model — the most accurate, neuroaffirming framing [10].

Avoid or re-label:

- Named **levels / bands** ("Low/Moderate/High/Extreme") — not validated; use position-on-dimension + profile instead.
- **"Attentional inertia"** as an umbrella — it blurs autistic inertia (task/control) with sticky attention (perception); use the precise term [3][6].
- **"Absolute" monotropism** and **"neurological shock"** — not recognised terms; imply a categorical jump and a pathology the data don't support.
- Stating the **monotropism** → **burnout** pathway as established — it is a strong hypothesis, not a measured effect [8].

- Any framing where **higher** = **more disordered** — monotropism's outcomes depend on environmental fit, not level [1][10].

Sources

[1] Murray, D., Lesser, M., & Lawson, W. (2005). *Attention, monotropism and the diagnostic criteria for autism.* Autism, 9(2), 139–156. doi:10.1177/1362361305051398 — origin of monotropism theory; frames it as a difference in resource allocation (a continuum), not a deficit. https://doi.org/10.1177/1362361305051398

[2] Garau, V., Woods, R., Chown, N., Hallett, S., Murray, F., Wood, R., Murray, A. & Fletcher-Watson, S. (2023). *The Monotropism Questionnaire* (47-item self-report measure; validation preprint). Open Science Framework. — dimensional 1–5 score + 8 subscales; n=1,110; autistic mean 4.15 vs non-autistic 3.19; ADHD raises scores; **no cutoff/percentile band/severity level** (confirmed by scanning the validation manuscript). Questionnaire: https://osf.io/wpx5g · validation preprint: https://osf.io/ft73y · CC BY-NC-SA 4.0. Local archive: `sources/17-monotropism-questionnaire-osf.pdf`

[3] Buckle, K. et al. (2021). *"No way out except from external intervention": First-hand accounts of autistic inertia.* (Preprint / qualitative). doi:10.31234/osf.io/ahk6x — autistic people describe difficulty starting, stopping, and changing activities "that was not within their conscious control." Local archive: `sources/11-autistic-inertia-osf.pdf` · https://doi.org/10.31234/osf.io/ahk6x · Manchester Research Explorer: https://research.manchester.ac.uk/en/publications/no-way-out-except-from-external-intervention-first-hand-accounts-

[4] Srinivasan, D. (2025). *The Physics of Autistic Inertia* (Student Notebook). APS Observer, Association for Psychological Science. — frames autistic inertia as "first articulated in autistic community narratives," now peer-reviewed; uses "high-friction launch" and "switching costs" language; cites Carls-Diamante & Laciny (2024), Greaves-Lord et al. (2023), Buckle et al. (2021), Heasman et al. (2024), Rapaport et al. (2024), Phung et al. (2021). https://www.psychologicalscience.org/publications/observer/student-notebook-autistic-inertia-srinivasan.html

[5] Greaves-Lord, L. et al. (2023). *[Getting 'stuck': an integrative approach to support autistic people]* Frontiers in Psychology, doi:10.3389/fpsyg.2023.1229596. — peer-reviewed qualitative + Delphi framework on limited response initiation ("stuck state") in autistic people across micro/meso/macro levels, co-created with autistic people. https://doi.org/10.3389/fpsyg.2023.1229596

[6] Landry, R. & Bryson, S. (2004). *Impaired disengagement of attention in young children with autism.* Journal of Child Psychology and Psychiatry, 45(6), 1115–1122. doi:10.1111/j.1469-7610.2004.00304.x — "sticky attention": slowed disengagement from a stimulus. https://doi.org/10.1111/j.1469-7610.2004.00304.x

[7] Sacrey, L.-A. et al. (2015). *Attention disengagement in autism spectrum disorder: A meta-analysis.* Neuroscience & Biobehavioral Reviews, 56, 111–121. doi:10.1016/j.neubiorev.2014.10.011 — confirms slowed attention disengagement across studies. https://doi.org/10.1016/j.neubiorev.2014.10.011

[8] Raymaker, D. M. et al. (2020). *"Having all of your internal resources exhausted beyond measure and being left with no clean-up crew": Defining autistic burnout.* Autism in Adulthood, 2(2), 132–143. doi:10.1089/aut.2019.0079 — defines autistic burnout as chronic exhaustion, loss of skills, reduced tolerance to stimulus, from chronic life stress and a mismatch of expectations and abilities. https://journals.sagepub.com/doi/abs/10.1089/aut.2019.0079 · open archive: https://pdxscholar.library.pdx.edu/socwork_fac/378

[9] Higgins, J. M. et al. (2021). *Defining autistic burnout through experts by lived experience: Grounded Delphi method investigating #AutisticBurnout.* Autism, 25(8). doi:10.1177/13623613211019858 — consensus definition: exhaustion, withdrawal, problems with thinking, reduced daily living skills, increased autistic traits. https://www.autismcrc.com.au/knowledge-centre/publications/defining-autistic-burnout-using-grounded-delphi-method-autistic

[10] Goldknopf, E. J. (2013). *Atypical resource allocation may contribute to many aspects of autism.* Frontiers in Integrative Neuroscience. PMC3872719 — models autism as a difference in *resource allocation* (narrowed and/or reduced attentional resources); the academic form of "cognitive energy economics." https://pmc.ncbi.nlm.nih.gov/articles/PMC3872719

[11] Rapaport, H., Clapham, H., Adams, J., Lawson, W., Porayska-Pomsta, K., & Pellicano, E. (2024). *"In a State of Flow": experiential validity of monotropism and its relation to autistic wellbeing.* Autism in Adulthood. doi:10.1089/aut.2023.0032 — the positive pole of deep monotropic immersion (flow); supports the "asset when supported" framing. Local archive: `sources/18-flow-experiential-validity-pmc.html` · https://doi.org/10.1089/aut.2023.0032